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1. Introduction

2. Model

3. Introduction and model formulation

3.1. **Physical origin and governing equations.** The transition from laminar to turbulent pipe flow has been investigated by Barkley and collaborators using a reduced model coupling turbulence intensity to local centerline velocity. The original program was initiated in [1] and developed into the model used to describe the rise of fully turbulent flow in [3, 2, 4].

The spatially homogeneous reaction dynamics associated with the later model are

$$\begin{aligned} \dot{q} &= q\big[r + u - 2 - (r + 0.1)(q - 1)^2\big], \\ \dot{u} &= \varepsilon\big[2 - u + 2q(1 - u)\big], \end{aligned} \tag{1}$$

where $q \geq 0$ represents turbulence intensity, u is the centerline-velocity variable, $r > 0$ is a Reynolds-number-related model parameter, and $\varepsilon > 0$ is a timescale parameter. Equations (3) arise by suppressing the spatial-derivative terms in the published reaction–diffusion–advection model; they are not a new artificial modification [3].

4. Model formulation

Barkley's reduced description of transitional pipe flow couples turbulence intensity to the local centerline velocity [1, 3, 2]. Suppressing the spatial-derivative terms gives the homogeneous reaction system

$$\begin{aligned} \dot{q} &= q\left[r + u - 2 - (r + \delta)(q - 1)^2\right], \\ \dot{u} &= \varepsilon\left[2 - u + \kappa q(1 - u)\right], \end{aligned} \qquad r, \delta, \varepsilon, \kappa > 0. \tag{2}$$

Here $q \geq 0$ measures turbulence intensity, u denotes the normalized centerline velocity, and r is a Reynolds-number-related control parameter. The velocity normalization assigns the value $u = 2$ to the laminar state, while the turbulence-dependent term drives u toward the reference level $u = 1$. The parameter ε controls the relative timescale of velocity relaxation, and κ measures the strength of its coupling to turbulence.

The parameter δ controls the linear decay of turbulence near the laminar equilibrium $(q, u) = (0, 2)$: the corresponding eigenvalues are $-\delta$ and $-\varepsilon$. Thus $\delta > 0$ preserves the local stability of the laminar state, while $\kappa > 0$ represents the reduction of centerline velocity associated with increasing turbulence.

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In this investigation, we adopt the parameter choices $\delta = 0.1$ and $\kappa = 2$, obtaining

$$\begin{aligned} \dot{q} &= q\big[r + u - 2 - (r + 0.1)(q - 1)^2\big], \\ \dot{u} &= \varepsilon\big[2 - u + 2q(1 - u)\big], \end{aligned} \qquad r, \varepsilon > 0. \tag{3}$$

These choices fix the laminar decay rate and the turbulence–velocity coupling strength, leaving r and ε as the bifurcation parameters. System (3) is a cubic polynomial differential system. Our aim is to investigate the existence, stability, and number of its limit cycles in the physical half-plane $q \geq 0$. Such cycles describe spatially homogeneous temporal oscillations of the reduced model; they should be distinguished from traveling or spatially localized structures of the full pipe-flow equations.

The nonzero equilibria satisfy

$$2(r + 0.1)q^3 - 3(r + 0.1)q^2 + 2(1 - r)q + 0.1 = 0.$$

Dividing by $2(r + 0.1)$ and substituting $q = z + \frac{1}{2}$ reduces this algebraic equation to

$$z^3 + p(r)z + b(r) = 0,$$

where

$$p(r) = \frac{37 - 70r}{40(r + 0.1)}, \qquad b(r) = \frac{21 - 30r}{40(r + 0.1)}.$$

This substitution is used only to solve the equilibrium equation; it does not change the variables of the differential system.

For $r > r_c \approx 0.645189835$, define

$$s(r) = \sqrt{\frac{70r - 37}{120(r + 0.1)}}, \qquad \theta(r) = \frac{1}{3}\arccos\left(\frac{30r - 21}{80(r + 0.1)s(r)^3}\right). \tag{4}$$

Here arccos takes its principal value in $[0, \pi]$. The two positive roots are explicitly

$$\boxed{\begin{aligned} q_-(r) &= \frac{1}{2} + 2s(r)\cos\left(\theta(r) - \frac{2\pi}{3}\right), \\ q_+(r) &= \frac{1}{2} + 2s(r)\cos\theta(r). \end{aligned}} \tag{5}$$

They satisfy

$$0 < q_-(r) < q_+(r) < 2.$$

Consequently, the two positive equilibria are

$$\boxed{E_\pm(r) = \left(q_\pm(r), \frac{2[1 + q_\pm(r)]}{1 + 2q_\pm(r)}\right).} \tag{6}$$

The remaining root is

$$q_{\mathrm{n}}(r) = \frac{1}{2} + 2s(r)\cos\left(\theta(r) + \frac{2\pi}{3}\right),$$

and satisfies $-\frac{1}{2} < q_{\mathrm{n}}(r) < 0$. Its equilibrium is outside the physical half-plane.

At $r = r_c$, the same formulas hold by continuity: $\theta(r_c) = \pi/3$ and

$$q_-(r_c) = q_+(r_c) = q_c.$$

For $0 < r < r_c$, there are no positive roots.

For example, at $r = 0.65$ these exact expressions give

$$\begin{aligned} E_- &\approx (0.708427852,\ 1.413760738), \\ E_+ &\approx (0.896537216,\ 1.358028411). \end{aligned}$$

Together with $E_0 = (0, 2)$, these are all physical equilibria for this parameter value.

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4.1. **Equilibria and conditions for limit-cycle existence.** Consider

$$\begin{aligned} \dot{q} &= q\left[r + u - 2 - (r+0.1)(q-1)^2\right], \\ \dot{u} &= \varepsilon\left[2 - u + 2q(1-u)\right], \end{aligned} \qquad r, \varepsilon > 0. \tag{7}$$

Equilibria. Setting $\dot{q} = \dot{u} = 0$ gives the laminar equilibrium

$$E_0 = (0, 2).$$

Every other equilibrium has the form

$$E_* = \left(q_*, \frac{2(1+q_*)}{1+2q_*}\right), \tag{8}$$

where q_* is a real root of

$$2(r+0.1)q^3 - 3(r+0.1)q^2 + 2(1-r)q + 0.1 = 0. \tag{9}$$

Thus (9), together with (8), specifies all nonzero equilibria exactly.

There is always exactly one negative root $q_{\mathrm{n}} \in (-1/2, 0)$. Its equilibrium lies outside the physical region $q \geq 0$.

To count the positive roots, let q_c be the unique positive root of

$$44q_c^3 - 30q_c^2 - 3q_c - 1 = 0,$$

and define

$$r_c = \frac{1 + 22q_c}{10q_c(2-q_c)(1+2q_c)} - \frac{1}{10}.$$

Numerically,

$$q_c \approx 0.802140182, \qquad r_c \approx 0.645189835.$$

The physical equilibria are therefore:

Parameter range	Equilibria in $q \geq 0$
$0 < r < r_c$	E_0
$r = r_c$	$E_0,\ E_c$ (a double positive equilibrium)
$r > r_c$	$E_0,\ E_-,\ E_+$

where

$$E_\pm = \left(q_\pm, \frac{2(1+q_\pm)}{1+2q_\pm}\right), \qquad 0 < q_- < q_+ < 2.$$

The equilibrium coordinates do not depend on ε.

Linear stability. At E_0, the eigenvalues are

$$\lambda_1 = -0.1, \qquad \lambda_2 = -\varepsilon,$$

so the laminar equilibrium is always locally asymptotically stable.

At a nonzero equilibrium, the Jacobian is

$$J_* = \begin{pmatrix} 2(r+0.1)q_*(1-q_*) & q_* \\ -\dfrac{2\varepsilon}{1+2q_*} & -\varepsilon(1+2q_*) \end{pmatrix}. \tag{10}$$

Its trace and determinant are

$$T_* = 2(r+0.1)q_*(1-q_*) - \varepsilon(1+2q_*), \tag{11}$$

$$\Delta_* = 2\varepsilon q_* \left[(r+0.1)(q_*-1)(1+2q_*) + \frac{1}{1+2q_*}\right]. \tag{12}$$

The negative equilibrium and the lower positive equilibrium E_- are saddles. At the upper equilibrium E_+, one has $\Delta_+ > 0$; hence it is asymptotically stable when $T_+ < 0$ and unstable when $T_+ > 0$.

Test for a Hopf bifurcation. The relevant equilibrium for the birth of small limit cycles is E_+. The necessary spectral conditions are

$$T_+ = 0, \qquad \Delta_+ > 0.$$

The trace condition gives

$$\boxed{\varepsilon_H = \frac{2(r+0.1)q_+(1-q_+)}{1+2q_+}.} \tag{13}$$

This threshold is positive precisely when

$$\boxed{r_c < r < \frac{2}{3}.}$$

For each fixed r in this interval, $\Delta_+ > 0$ at $\varepsilon = \varepsilon_H$, and

$$\left.\frac{d\ \mathrm{Re}\,\lambda}{d\varepsilon}\right|_{\varepsilon=\varepsilon_H} = -\frac{1+2q_+}{2} \neq 0.$$

Thus the spectral and transversality conditions hold.

To conclude that a nondegenerate Hopf bifurcation actually produces a limit cycle, one must additionally verify that the first Lyapunov coefficient is nonzero. Its sign determines the stability of the cycle and the side of ε_H on which it exists. The condition $T_+ = 0$ alone is not sufficient.

For example, when $r = 0.65$,

$$q_+ \approx 0.896537216, \qquad u_+ \approx 1.358028411, \qquad \varepsilon_H \approx 0.049815126.$$

This is a concrete parameter point at which to perform the Hopf nondegeneracy test.
A direct nonexistence test. For the vector field (P, Q) in (7), the Dulac function $B(q, u) = 1/q$ on $q > 0$ gives

$$\frac{\partial(BP)}{\partial q} + \frac{\partial(BQ)}{\partial u} = 2(r+0.1)(1-q) - \frac{\varepsilon(1+2q)}{q}.$$

This expression is nonpositive throughout $q > 0$ if

$$\boxed{\varepsilon \geq (2-\sqrt{3})(r+0.1).}$$

The Bendixson–Dulac criterion then excludes all physical limit cycles. There are also no physical limit cycles for $r \leq r_c$.

Therefore, the Hopf test identifies where small cycles can arise, while the Dulac test excludes a region of parameter space. Neither test by itself determines the global number of limit cycles.

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5. Preliminary analysis of limit cycles

Throughout this section, let

$$A = r + \frac{1}{10}, \qquad h(q) = \frac{21}{10} - 2Aq + Aq^2, \tag{14}$$

and define the polynomials

$$\begin{aligned} H(q) &= (1+2q)h(q) - 2 - 2q \\ &= 2Aq^3 - 3Aq^2 + \left(\frac{11}{5} - 2A\right)q + \frac{1}{10}, \\ D(q) &= \varepsilon(1+2q) + qh'(q) \\ &= 2Aq^2 + 2(\varepsilon - A)q + \varepsilon. \end{aligned} \tag{15}$$

The original equations can then be written as

$$\dot{q} = q[u - h(q)], \qquad \dot{u} = \varepsilon[2 + 2q - (1 + 2q)u]. \tag{16}$$

5.1. **Invariant region and boundedness.** The half-plane $q \geq 0$ is positively invariant because $q = 0$ is an invariant line. Moreover, every solution with $q(0) > 0$ remains in $q > 0$ throughout its interval of existence.

The strip

$$\mathcal{S} = \{(q, u) : q \geq 0,\ 1 \leq u \leq 2\} \tag{17}$$

is positively invariant. Indeed,

$$\dot{u}|_{u=1} = \varepsilon > 0, \qquad \dot{u}|_{u=2} = -2\varepsilon q \leq 0.$$

More generally, if $q(0) \geq 0$, comparison at the horizontal boundaries gives

$$\min\{u(0), 1\} \leq u(t) \leq \max\{u(0), 2\}.$$

Since u is bounded and the leading term of $\dot{q}$ is $-Aq^3$, the component q also remains bounded. Consequently, every solution starting in $q \geq 0$ exists for all forward time.

Every nonconstant periodic orbit in the physical half-plane lies in

$$q > 0, \qquad 1 < u < 2. \tag{18}$$

To see this, first note that a periodic orbit cannot intersect the invariant line $q = 0$, on which $\dot{u} = \varepsilon(2 - u)$. At a maximum or minimum of u on a periodic orbit,

$$u = \frac{2(1 + q)}{1 + 2q} = 1 + \frac{1}{1 + 2q},$$

which belongs to $(1, 2)$ when $q > 0$.

5.2. **Equilibria and their stability.** The laminar equilibrium

$$E_0 = (0, 2)$$

exists for all $r, \varepsilon > 0$. Its Jacobian is

$$J(E_0) = \begin{pmatrix} -\frac{1}{10} & 0 \\ -2\varepsilon & -\varepsilon \end{pmatrix},$$

so E_0 is locally asymptotically stable.

A positive equilibrium $E_* = (q_*, u_*)$ satisfies

$$u_* = \frac{2(1 + q_*)}{1 + 2q_*}, \qquad H(q_*) = 0. \tag{19}$$

Equivalently, the positive-equilibrium branch is parameterized by

$$r = R(q) := \frac{1 + 22q}{10q(2 - q)(1 + 2q)} - \frac{1}{10}, \qquad 0 < q < 2. \tag{20}$$

Differentiation yields

$$R'(q) = \frac{C(q)}{5q^2(q - 2)^2(2q + 1)^2}, \qquad C(q) = 44q^3 - 30q^2 - 3q - 1. \tag{21}$$

By Descartes' rule of signs, C has exactly one positive root. Denote it by q_{SN}. Since $C(0) < 0 < C(1)$, this root lies in $(0, 1)$. Thus R decreases on $(0, q_{\mathrm{SN}})$ and increases on $(q_{\mathrm{SN}}, 2)$, with $R(q) \to +\infty$ at both endpoints.

Define

$$r_{\mathrm{SN}} = R(q_{\mathrm{SN}}). \tag{22}$$

Numerically,

$$q_{\mathrm{SN}} \approx 0.802140182, \qquad r_{\mathrm{SN}} \approx 0.645189835.$$

It follows that there are no positive equilibria for $r < r_{\mathrm{SN}}$, one double positive equilibrium for $r = r_{\mathrm{SN}}$, and two positive equilibria

$$E_{\pm} = \left(q_{\pm}, \frac{2(1+q_{\pm})}{1+2q_{\pm}} \right), \qquad q_{-} < q_{+},$$

for $r > r_{\mathrm{SN}}$.

At a positive equilibrium, the Jacobian is

$$J_* = \begin{pmatrix} 2Aq_*(1-q_*) & q_* \\ -\dfrac{2\varepsilon}{1+2q_*} & -\varepsilon(1+2q_*) \end{pmatrix}. \tag{23}$$

In particular,

$$\operatorname{tr} J_* = -D(q_*), \qquad \det J_* = \varepsilon q_* H'(q_*). \tag{24}$$

For $r > r_{\mathrm{SN}}$,

$$H'(q_-) < 0, \qquad H'(q_+) > 0.$$

Hence E_- is a saddle, whereas E_+ is asymptotically stable when $D(q_+) > 0$ and unstable when $D(q_+) < 0$.

Since $R(1) = 2/3$, the upper equilibrium satisfies

$$q_+ < 1 \quad \Longleftrightarrow \quad r_{\mathrm{SN}} < r < \frac{2}{3}.$$

For $r \geq 2/3$, one has $q_+ \geq 1$ and therefore $D(q_+) > 0$ for every $\varepsilon > 0$.

5.3. **Hopf bifurcation and local limit cycles.** A positive Hopf threshold exists precisely on the upper branch with $r_{\mathrm{SN}} < r < 2/3$. It is given by

$$\varepsilon_H(r) = \frac{2Aq_+(1-q_+)}{1+2q_+}. \tag{25}$$

At this threshold,

$$D(q_+) = 0, \qquad \omega^2 := \varepsilon_H q_+ H'(q_+) > 0.$$

The equilibrium coordinates are independent of ε, and

$$\left. \frac{\partial \operatorname{Re} \lambda}{\partial \varepsilon} \right|_{\varepsilon = \varepsilon_H} = -\frac{1+2q_+}{2} < 0. \tag{26}$$

Thus the transversality condition holds.

For the calculation of the Hopf coefficient, the polynomial substitution

$$v = u - h(q)$$

gives

$$\dot{q} = qv, \qquad \dot{v} = -D(q)v - \varepsilon H(q). \tag{27}$$

This auxiliary system remains cubic polynomial and uses the original time variable.

At the Hopf point, set $s = q_+$ and introduce the linear coordinates

$$X = q - s, \qquad Y = -\frac{s}{\omega} v.$$

The linear part becomes $\dot{X} = -\omega Y$, $\dot{Y} = \omega X$. The cubic coefficient in the corresponding radial normal form is

$$\ell = \frac{D'(s)H''(s) - D''(s)H'(s)}{16H'(s)}, \tag{28}$$

where D is evaluated at $\varepsilon = \varepsilon_H$. Using $H(s) = 0$ and (25), this simplifies to

$$\ell = \frac{(22s+1)\left(88s^4 + 176s^3 - 186s^2 + 64s + 5\right)}{80s(2-s)(2s+1)^2 C(s)}. \tag{29}$$

For $q_{\mathrm{SN}} < s < 1$, the denominator is positive. Moreover,

$$88s^4 + 176s^3 - 186s^2 + 64s + 5 = 88s^4 + s(176s^2 - 186s + 64) + 5 > 0,$$

because the quadratic $176s^2 - 186s + 64$ has positive leading coefficient and negative discriminant. Consequently,

$$\ell > 0. \tag{30}$$

Therefore, for every fixed $r \in (r_{\mathrm{SN}}, 2/3)$, the Hopf bifurcation is nondegenerate and subcritical. A unique small-amplitude unstable limit cycle bifurcates from E_+ for $\varepsilon > \varepsilon_H(r)$ sufficiently close to $\varepsilon_H(r)$. The equilibrium inside this cycle is asymptotically stable.

This uniqueness is local: it concerns cycles sufficiently close to the Hopf equilibrium and does not exclude additional cycles elsewhere in the phase plane.

5.4. **Restrictions on the location of periodic orbits.** At an extremum of q on a nonconstant periodic orbit, $\dot{q} = 0$ and hence $u = h(q)$. Differentiating the first equation of (16) gives, at such a point,

$$\ddot{q} = -\varepsilon q H(q). \tag{31}$$

The extremum cannot satisfy $H(q) = 0$, since it would then be an equilibrium point.

If $r \le r_{\mathrm{SN}}$, one has $H(q) \ge 0$ for all $q > 0$. Equation (31) then excludes the minimum required by any nonconstant periodic orbit. Thus

$$0 < r \le r_{\mathrm{SN}} \quad \Longrightarrow \quad \text{no limit cycles in } q \ge 0. \tag{32}$$

Suppose now that $r > r_{\mathrm{SN}}$. Since H is negative precisely between its positive roots, the minimum and maximum of q along any periodic orbit satisfy

$$q_- < q_{\min} < q_+ < q_{\max} < 2. \tag{33}$$

For the final inequality, recall that $u < 2$ along the orbit, whereas

$$h(q) = \frac{21}{10} + Aq(q-2) \ge \frac{21}{10} \qquad \text{for } q \ge 2.$$

Hence a q-maximum cannot occur at $q \ge 2$.

In particular, every physical limit cycle excludes the saddle E_- and the boundary equilibrium E_0. By the index theorem for planar periodic orbits, it surrounds E_+.

5.5. **A Bendixson–Dulac exclusion region.** Let P and Q denote the right-hand sides of (3). On the simply connected half-plane $q > 0$, choose the Dulac function

$$B(q,u) = \frac{1}{q}.$$

A direct calculation gives

$$\frac{\partial(BP)}{\partial q} + \frac{\partial(BQ)}{\partial u} = 2A(1-q) - \frac{\varepsilon(1+2q)}{q} = -\frac{D(q)}{q}. \tag{34}$$

The quadratic polynomial D is nonnegative for all $q > 0$ if

$$\varepsilon \ge (2-\sqrt{3})A. \tag{35}$$

Indeed, for $\varepsilon \ge A$ all its coefficients are nonnegative. For $0 < \varepsilon < A$, its minimum is

$$\min_{q>0} D(q) = \varepsilon - \frac{(A-\varepsilon)^2}{2A},$$

which is nonnegative precisely when $\varepsilon \ge (2-\sqrt{3})A$.

The Bendixson–Dulac criterion therefore yields

$$\boxed{\varepsilon \ge (2-\sqrt{3})\left(r + \frac{1}{10}\right) \quad \Longrightarrow \quad \text{no limit cycles in } q \ge 0.} \tag{36}$$

At equality, the Dulac divergence vanishes only on one vertical line, which does not affect the conclusion.

5.6. **Consequences for the global cycle problem.** The preceding results restrict the search for physical limit cycles to

$$r > r_{\mathrm{SN}}, \qquad 0 < \varepsilon < (2 - \sqrt{3}) \left(r + \frac{1}{10} \right).$$

Every such cycle surrounds the upper positive equilibrium and remains in the region specified by (18) and (33). Furthermore, a small unstable cycle is known to exist on the subcritical side of the Hopf curve (25).

These conclusions establish local existence and explicit global exclusion regions. They do not yet determine the total number of limit cycles or the global continuation of the Hopf branch. In particular, uniqueness away from the Hopf point and possible termination in a saddle homoclinic loop require additional analysis.

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